5.4 Triangle Proportionality

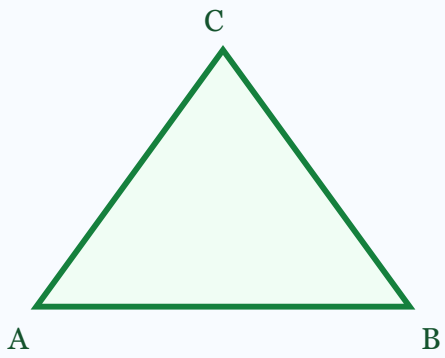
The **Triangle Proportionality Theorem** (Side-Splitter Theorem) says: if a line parallel to one side of a triangle intersects the other two sides, it divides those sides proportionally. For $\overline{DE} \parallel \overline{BC}$ with D on $\overline{AB}$ and E on $\overline{AC}$, this gives $\frac{AD}{DB} = \frac{AE}{EC}$, or equivalently the "whole-to-part" form $\frac{AD}{AB} = \frac{AE}{AC}$. The **converse** is also true: if a line divides two sides proportionally, it must be parallel to the third side. A useful corollary extends this to **three or more parallel lines**: they cut any two transversals into proportional segments. A special case is the **Midsegment Theorem**: the segment joining the midpoints of two sides is parallel to the third side and exactly half its length. This worksheet applies these ideas numerically, algebraically, and with a coordinate proof.

Examples

Example 1: Find a segment (part-to-part form)

In $\triangle ABC$, $\overline{DE} \parallel \overline{BC}$ with D on $\overline{AB}$ and E on $\overline{AC}$. If $AD = 4$, $DB = 6$, $AE = 6$, find EC.

Solution



$\overline{DE}$ parallel to the base splits both other sides proportionally.

Identify the theorem: since $\overline{DE} \parallel \overline{BC}$, $\frac{AD}{DB} = \frac{AE}{EC}$.

Substitute: $\frac{4}{6} = \frac{6}{EC}$

Cross-multiply: $4 \cdot EC = 36$

$EC = 9$.

Answer: $EC = 9$.

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Example 2: Find a segment (whole-to-part form)

$\overline{DE} \parallel \overline{BC}, AD = 3, AB = 8, AE = 6$. Find AC.

Solution

Use the whole-to-part form: $\dfrac{AD}{AB} = \dfrac{AE}{AC}$

$\dfrac{3}{8} = \dfrac{6}{AC}$

Cross-multiply: $3 \cdot AC = 48$

$AC = 16$.

Answer: $AC = 16$.

Example 3: Converse — test for parallel

In $\triangle ABC, AD = 5, DB = 10, AE = 4, EC = 8$. Is $\overline{DE} \parallel \overline{BC}$?

Solution

Compute both ratios: $\dfrac{AD}{DB} = \dfrac{5}{10} = \dfrac{1}{2}$ and $\dfrac{AE}{EC} = \dfrac{4}{8} = \dfrac{1}{2}$.

The ratios are equal.

Conclusion: Yes — by the Converse of the Triangle Proportionality Theorem, $\overline{DE} \parallel \overline{BC}$.

Example 4: Converse — not parallel

$AD = 3, DB = 5, AE = 4, EC = 6$. Is $\overline{DE} \parallel \overline{BC}$?

Solution

Compute both ratios: $\dfrac{AD}{DB} = \dfrac{3}{5} = 0.6$ and $\dfrac{AE}{EC} = \dfrac{4}{6} \approx 0.667$.

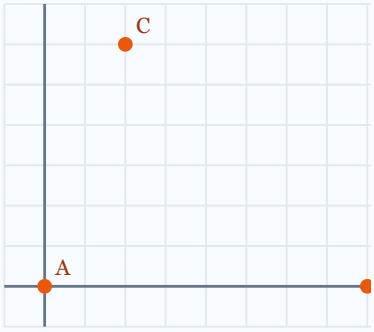
The ratios are not equal.

Conclusion: No — $\overline{DE}$ is not parallel to $\overline{BC}$.

Example 5: Midsegment — coordinate proof

Triangle has vertices $A(0,0)$, $B(8,0)$, $C(2,6)$. Let M be the midpoint of $\overline{AB}$ and N be the midpoint of $\overline{AC}$. Show $\overline{MN} \parallel \overline{BC}$ and $MN = \frac{1}{2}BC$.

Solution



M and N are the midpoints of $\overline{AB}$ and $\overline{AC}$.

Find the midpoints: $M = (\frac{0+8}{2}, 0) = (4,0)$; $N = (\frac{0+2}{2}, \frac{0+6}{2}) = (1,3)$.

Slope of $\overline{BC}$: $\frac{6-0}{2-8} = \frac{6}{-6} = -1$. Slope of $\overline{MN}$: $\frac{3-0}{1-4} = \frac{3}{-3} = -1$ — equal, so $\overline{MN} \parallel \overline{BC}$.

Length of $\overline{BC} = \sqrt{6^2+6^2} = 6\sqrt{2}$; length of $\overline{MN} = \sqrt{3^2+3^2} = 3\sqrt{2}$.

$\frac{MN}{BC} = \frac{3\sqrt{2}}{6\sqrt{2}} = \frac{1}{2}$.

Conclusion: $\overline{MN} \parallel \overline{BC}$ and $MN = \frac{1}{2}BC$, confirming the Midsegment Theorem.

Example 6: Solve for $x

$\overline{DE} \parallel \overline{BC}$, $AD = x$, $DB = 6$, $AE = 4$, $EC = 8$. Find x.

Solution

Set up the proportion: $\frac{x}{6} = \frac{4}{8} = \frac{1}{2}$

Cross-multiply: $2x = 6$

$x = 3$.

Answer: $x = 3$.

Example 7: Three parallel lines corollary

Three parallel lines cut two transversals, creating segments 4 and 6 on the first transversal, and 6 and x on the second (matched in the same order). Find x.

Solution

By the corollary to the Triangle Proportionality Theorem, three or more parallel lines cut any two transversals proportionally:

$$\frac{4}{6} = \frac{6}{x}$$

Cross-multiply: $4x = 36$

$x = 9$.

Answer: $x = 9$.

Example 8: Midsegment length

The midsegment of a triangle joins the midpoints of two sides. If the third side is 18, find the midsegment's length, and state which side it is parallel to.

Solution

By the Midsegment Theorem, the midsegment is half the length of the third side and parallel to it:

$$\text{Midsegment} = \frac{1}{2} \times 18 = 9.$$

Answer: length 9, parallel to the third side (the one not touched by either midpoint).

Example 9: Proportion in a labeled triangle

$\overline{FG} \parallel \overline{XY}$ in $\triangle WXY$, with $WF = 9, FX = 3, WG = 12$. Find GY.

Solution

Set up the proportion: $\frac{WF}{FX} = \frac{WG}{GY}$

$$\frac{9}{3} = \frac{12}{GY}$$

$$3 = \frac{12}{GY} \Rightarrow GY = 4.$$

Answer: $GY = 4$.

Example 10: Algebra with two expressions

$\overline{DE} \parallel \overline{BC}, AD = 3x - 2, DB = x + 4, AE = 8, EC = 4$. Find x.

Solution

Set up the proportion: $\frac{3x-2}{x+4} = \frac{8}{4} = 2$

Cross-multiply: $3x - 2 = 2(x + 4)$

$3x - 2 = 2x + 8$

$x = 10$.

Check: $AD = 3(10) - 2 = 28, DB = 10 + 4 = 14$, ratio $\frac{28}{14} = 2$ ✓.

Answer: $x = 10$.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Parallel

$\overline{DE} \parallel \overline{BC}, AD = 6, DB = 4, AE = 9$. Find EC.

Answer: **Verification:**

Problem 2: Parallel

$\overline{DE} \parallel \overline{BC}, AD = 5, AB = 12, AE = 10$. Find AC.

Answer: **Verification:**

Problem 3: Converse

$AD = 8, DB = 4, AE = 10, EC = 5$. Is $\overline{DE} \parallel \overline{BC}$?

Answer: **Verification:**

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Problem 4: Converse

$AD = 2, DB = 3, AE = 4, EC = 5$. Parallel?

Answer: **Verification:**

Problem 5: Solve x

$\overline{DE} \parallel \overline{BC}, AD = x, DB = 10, AE = 6, EC = 15$. Find x.

Answer: **Verification:**

Problem 6: Midsegment

Third side is 22. Find the midsegment length.

Answer: **Verification:**

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Problem 7: Find segment

$\overline{DE} \parallel \overline{BC}, AD = 4, DB = 8, AC = 18$. Find AE.

Answer: **Verification:**

Problem 8: Proportion

$\overline{FG} \parallel \overline{BC}, AF = 3, FB = 9, AG = 2$. Find GC.

Answer: **Verification:**

Problem 9: Midsegment

Midsegment length is 7. Find the third side.

Answer: **Verification:**

Problem 10: Parallel

$AD = 7, AB = 21, AE = 5$. If $\overline{DE} \parallel \overline{BC}$, find AC.

Answer: **Verification:**

Problem 11: Converse

$AD = 9, DB = 6, AE = 12, EC = 8$. Parallel?

Answer: **Verification:**

Problem 12: Find EC

$\overline{DE} \parallel \overline{BC}, AD = 10, DB = 15, AE = 8$. Find EC.

Answer: **Verification:**

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Problem 13: Challenge — three parallel lines

Three parallel lines cut two transversals into segments $4, 6$ on the first, and $6, x$ on the second (matched in order). Find x.

Answer: **Verification:**

Problem 14: Challenge — coordinate midsegment

Triangle has vertices $A(0, 0)$, $B(10, 0)$, $C(4, 8)$. Find midpoints M (of $\overline{AB}$) and N (of $\overline{AC}$), the length MN, and verify $\overline{MN} \parallel \overline{BC}$ with $MN = \frac{1}{2}BC$.

Answer: **Verification:**

Problem 15: Challenge — algebra with a ratio

$\overline{DE} \parallel \overline{BC}$, $AD = 2x + 1$, $DB = x + 5$, $AE = 12$, $EC = 8$. Find x, then AD, DB, and AB.

Answer: **Verification:**

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✓ Answer Key

1. 6
2. 24
3. Yes; both ratios $= 2$
4. No; $\frac{2}{3} \neq \frac{4}{5}$
5. 4
6. 11
7. 6
8. 6
9. 14
10. 15
11. Yes; both ratios $= \frac{3}{2}$
12. 12
13. $x = 9$
14. $M(5, 0), N(2, 4)$; $MN = 5, BC = 10$; both have slope $-\frac{4}{3}$, and $MN = \frac{1}{2}BC$
15. $x = 13$; $AD = 27, DB = 18, AB = 45$

Great work — keep practising!